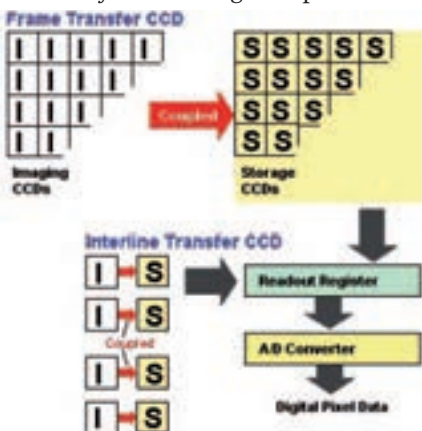


each pixel and translate and compile this into an image. The difference lies in how they go about doing it.

a) CCD sensors: That little P&S you've seen your friend or relative carrying most probably has a sensor commonly known as the CCD – an acronym for Charge-Coupled-Device.

What it's referring to is actually the technology that it uses to convert light to electricity. The reason why it's called 'coupled' is because of its interline architecture. The device is made of an array of imaging pixels and a corresponding array of storage pixels that are lined up or 'coupled' together. Once the imaging array is



Split second binaries

exposed to the light, it instantly transfers the charge to the storage array. This storage CCDs then transport the charge across the chip and it is 'read' by an ADC (Analog to Digital Converter) at one corner of the array. This converts the value (charge) of every pixel into a digital value (zeroes and ones) by gauging the intensity of light (by the amount of charge) at each photosite. The trouble that occurs here is that every alternate line is not an image sensor but a storage strip – opaque. So half of the chip's area is lost for the image. There's another sort of CCD, called the 'frame-transfer' CCD. half of the silicon area of the sensor is covered by an



The CMOS sensor - good things come in small packages